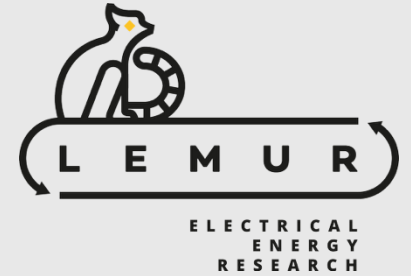




Universidad de Oviedo
Universidá d'Uviéu
University of Oviedo



Apagón - April 28, 2025

Update: May 31, 2025

Authors: Bryan Murray, Dr. Pablo García Fernández

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DISCLAIMERS



- The authors* believe there are perfectly plausible technical explanations for the Iberian blackout of April 28, 2025.
- The authors are focused on the technical explanations which can be justified with knowledge of the physical infrastructure. In other words, we are not considering cyber-attacks, human errors, or extraordinary meteorological conditions.
- For now, we are simply presenting what data we have and will use as a basis for much more engineering analysis.
- We are **NOT** presenting conclusions about root causes.
- Why? Because the authors currently lack critical data about the grid's operational conditions on April 28; for example, details about generators that were online but operating below rated power capacity.

*Report Authors

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Section 1

Overview of April 28

12.30

Según Red Eléctrica, el sistema tiene todas sus variables estables (frecuencia, tensión, flujo)

**12.33.16**

La red sufre un "evento" que parece una pérdida de generación. La frecuencia baja de golpe, pero la red se autoestabiliza



¿Qué ocasionó esas caídas?

+1,5 segundos

Se produce una segunda pérdida de generación. REE señala al suroeste y dice que es muy posible que la generación afectada sea solar



¿Por qué el sistema eléctrico no pudo reaccionar para compensar esa perturbación?

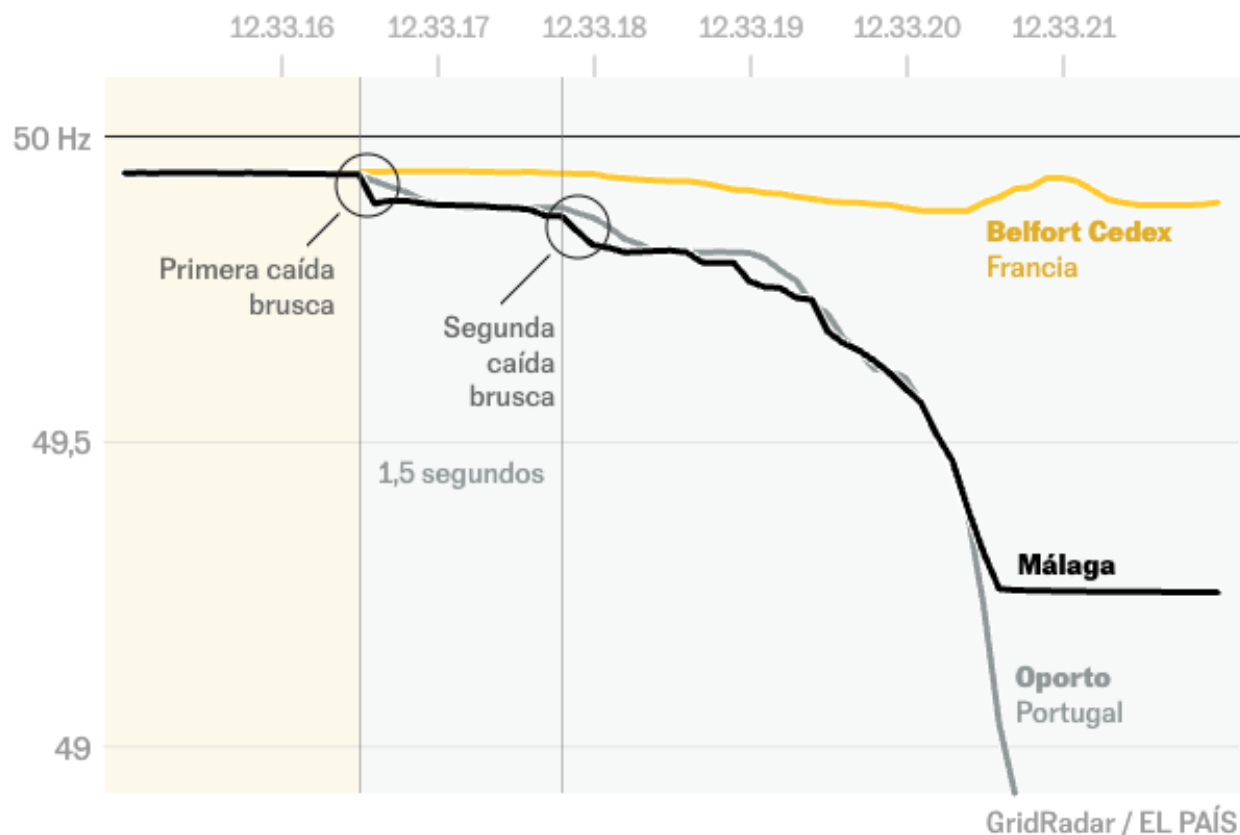
+3,5 segundos

La inestabilidad provoca la desconexión automática con Francia y Europa. Se precipita un colapso en cascada: los elementos de la red caen uno tras otro



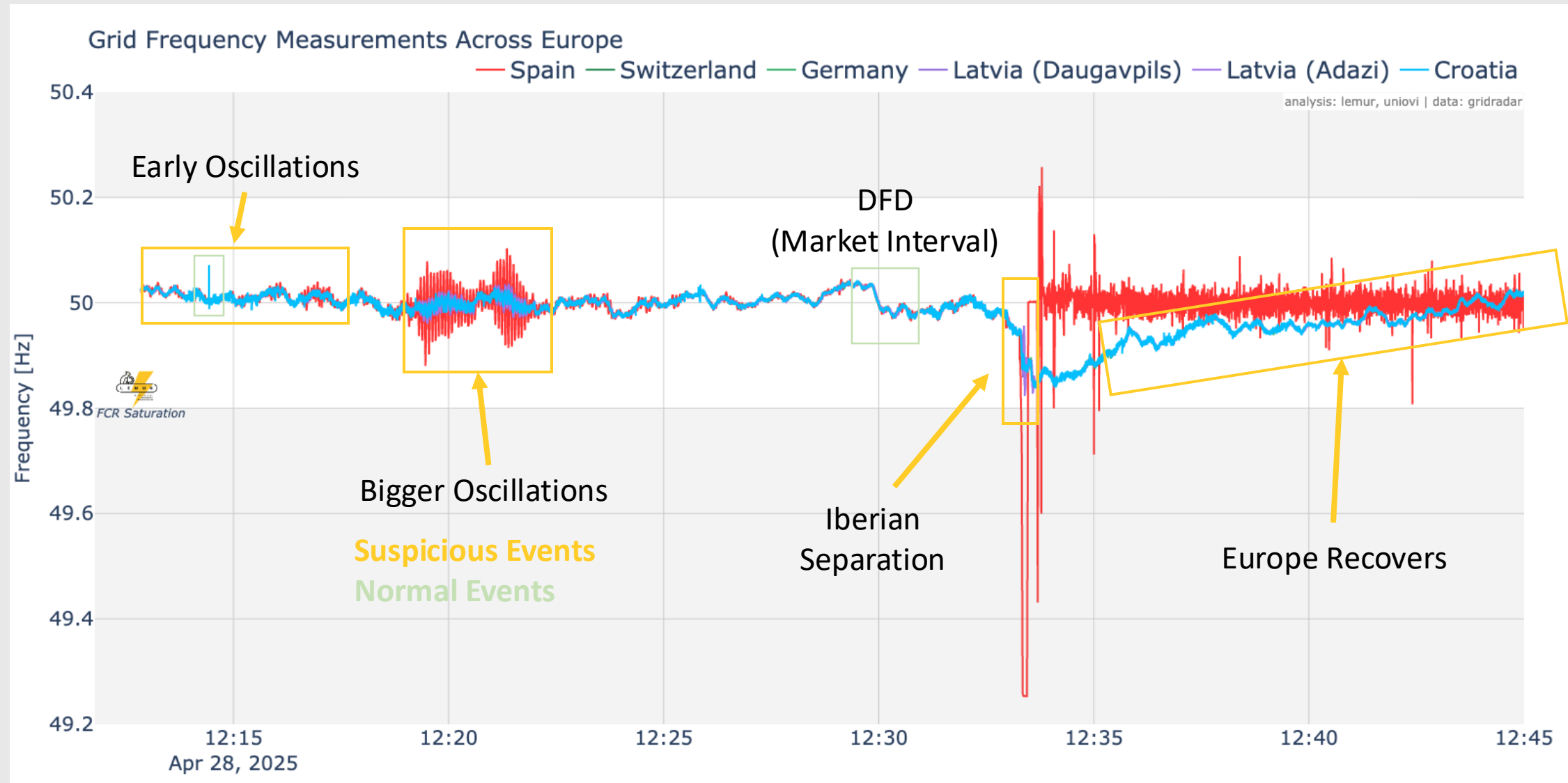
Fuentes: Red Eléctrica, Grid Radar y elaboración propia

Frecuencia de la red en Málaga, Oporto (Portugal) y Belfort (Francia)
Mediana cada 100 milisegundos

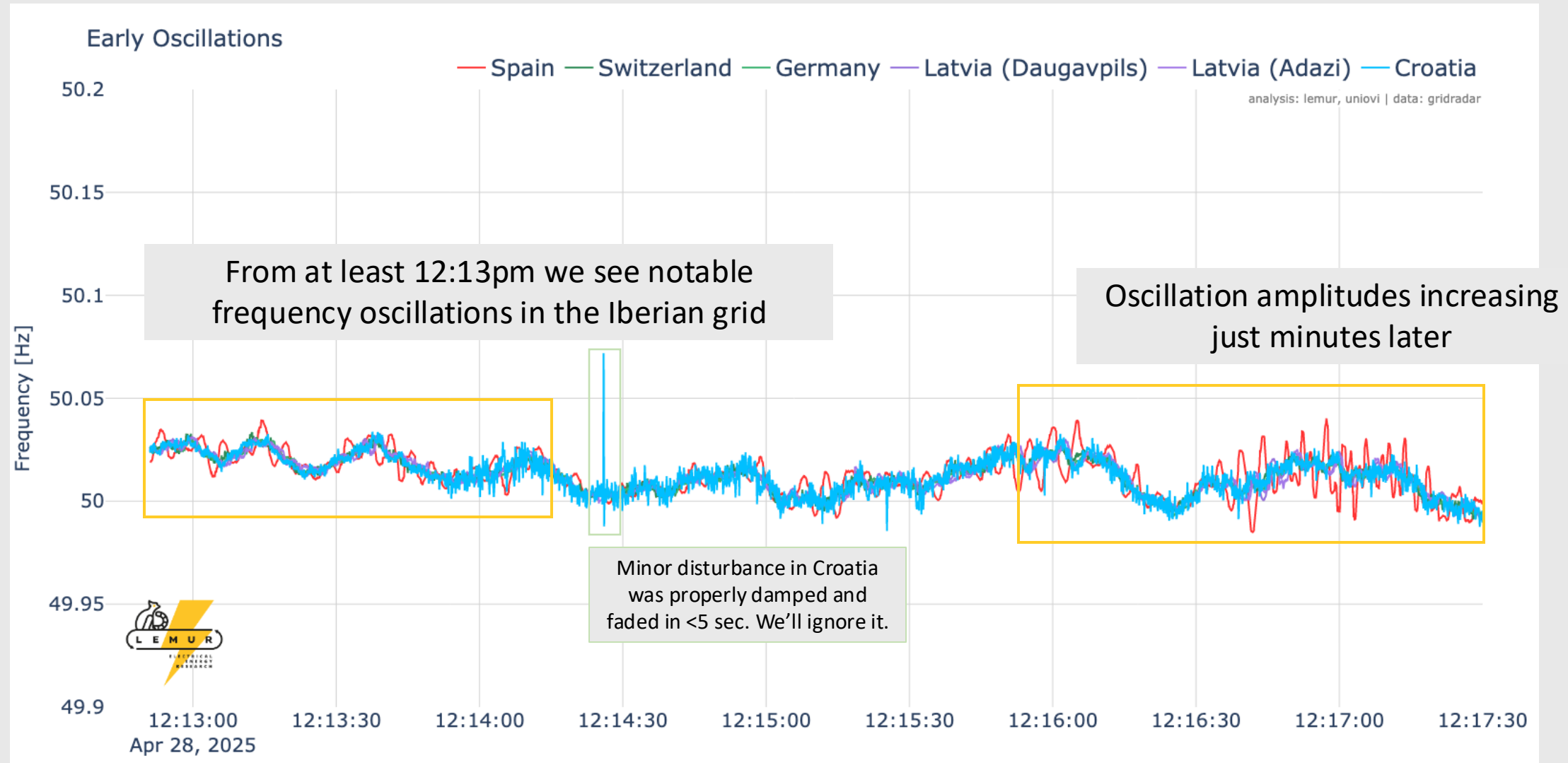




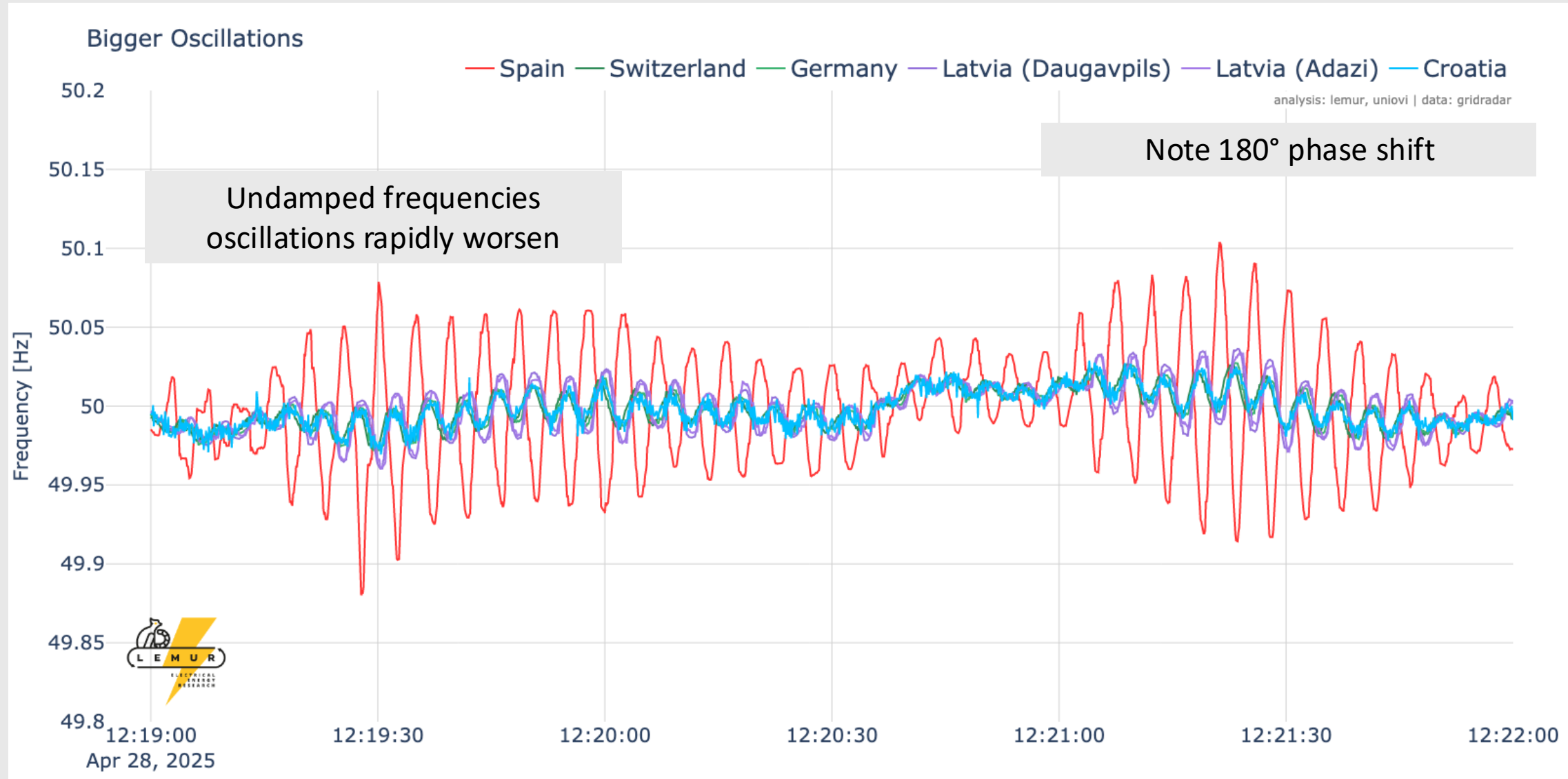
OVERVIEW OF EUROPEAN GRID FREQUENCIES JUST BEFORE THE BLACKOUT



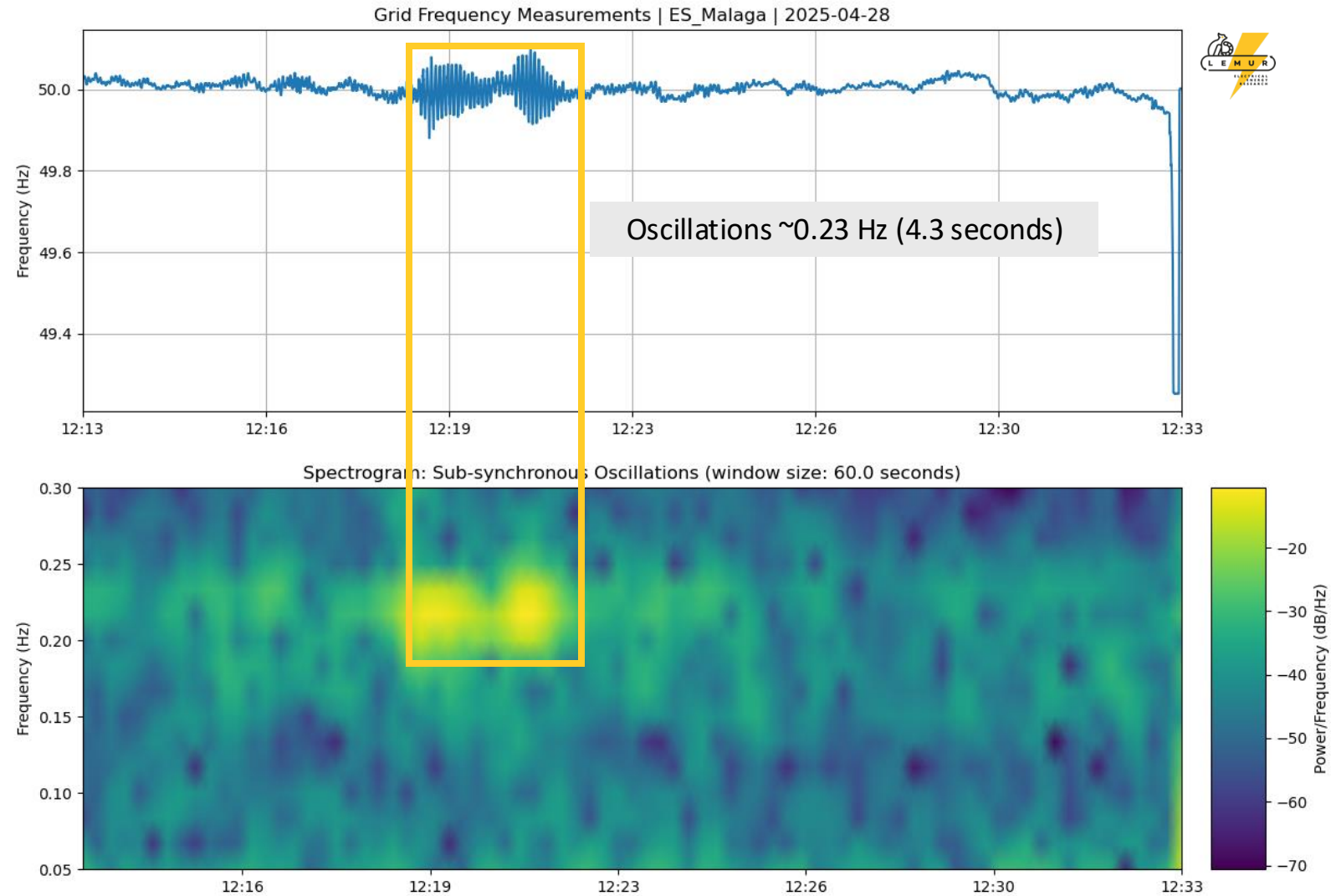
EARLY OSCILLATIONS HINT AT INSTABILITY



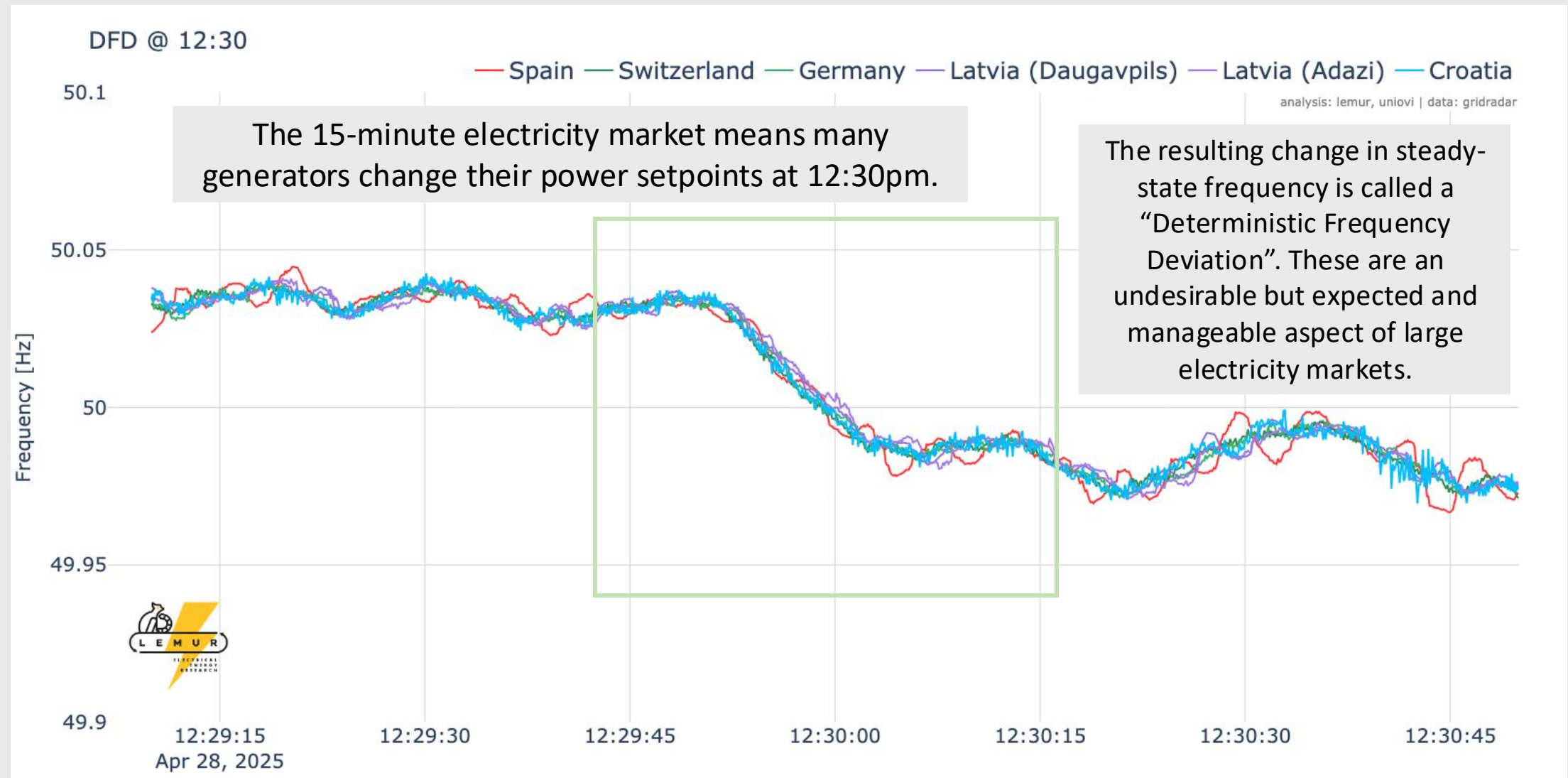
OSCILLATIONS WORSEN RAPIDLY ~ 12:20PM



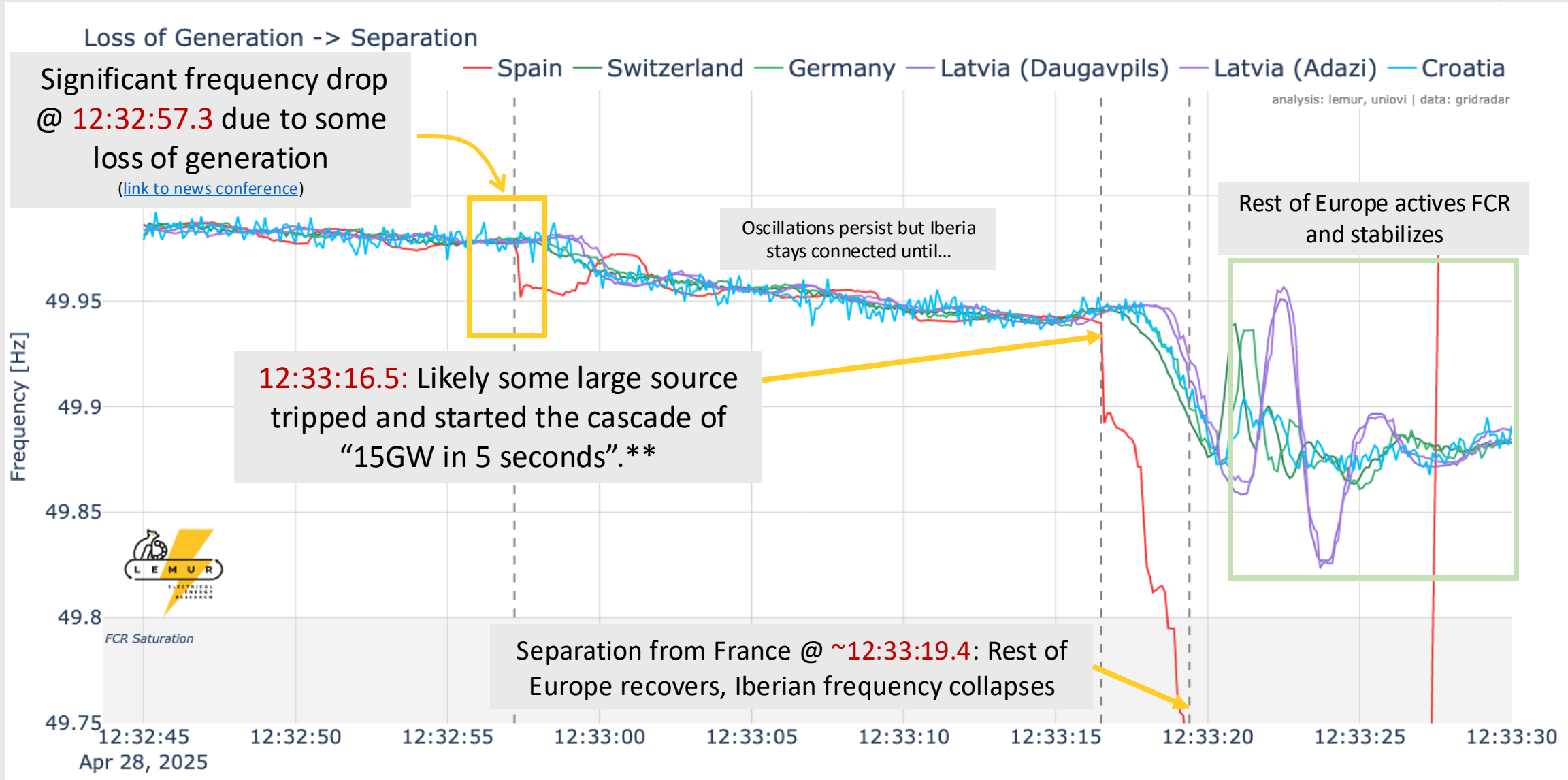
OBSERVING SPURIOUS OSCILLATIONS



MARKET CHANGE AT 12:30PM – NO PROBLEM

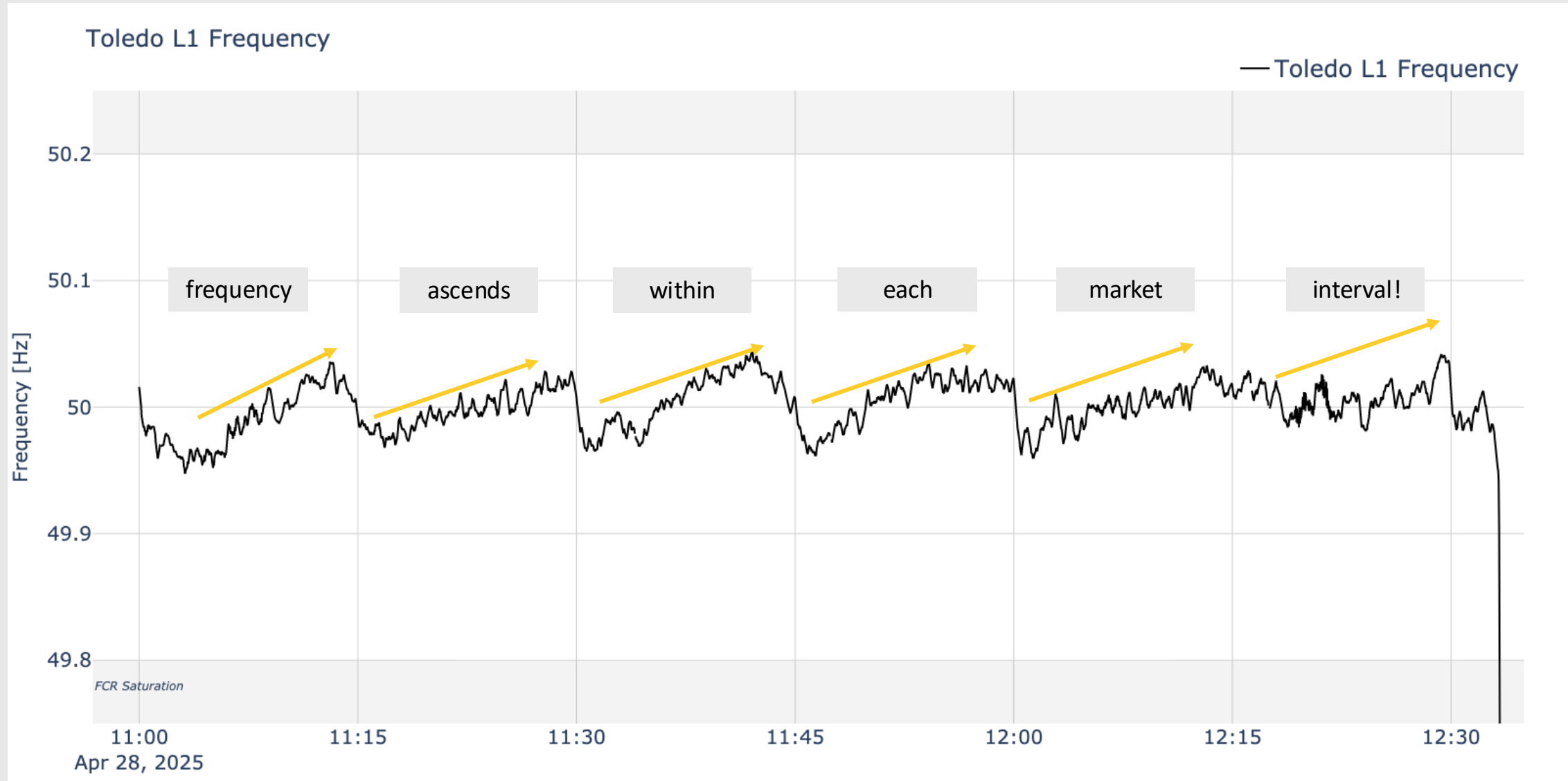


IBERIA SEPARATES FROM EUROPE



** "desconexión repentina de dos centrales de generación eléctrica en el suroeste de la península" [[El Periodico de la Energía, 29 Abril 2025](#)]

FREQUENCY IS CONSISTENTLY TO FAST ALL MORNING

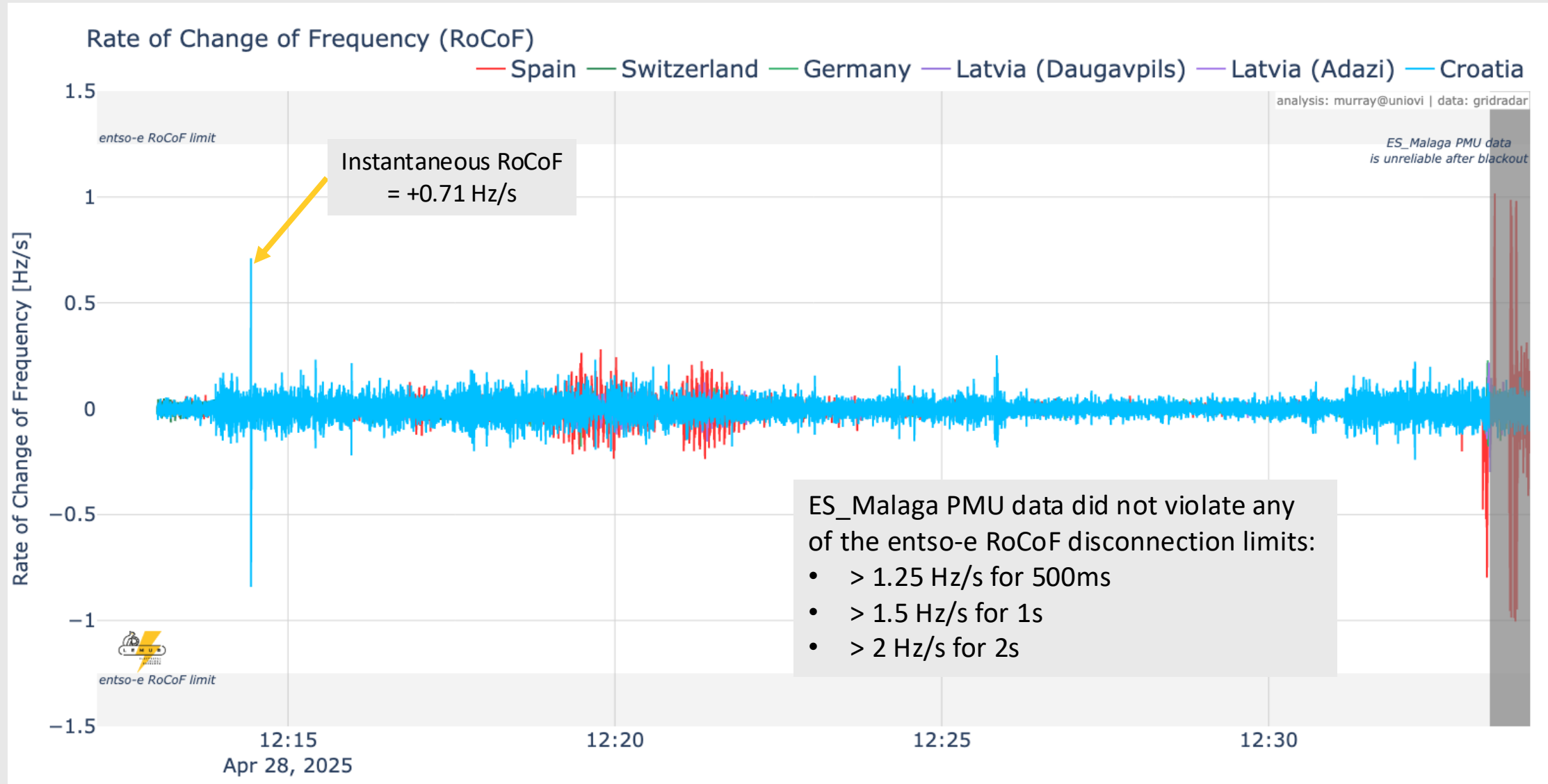




Section 2

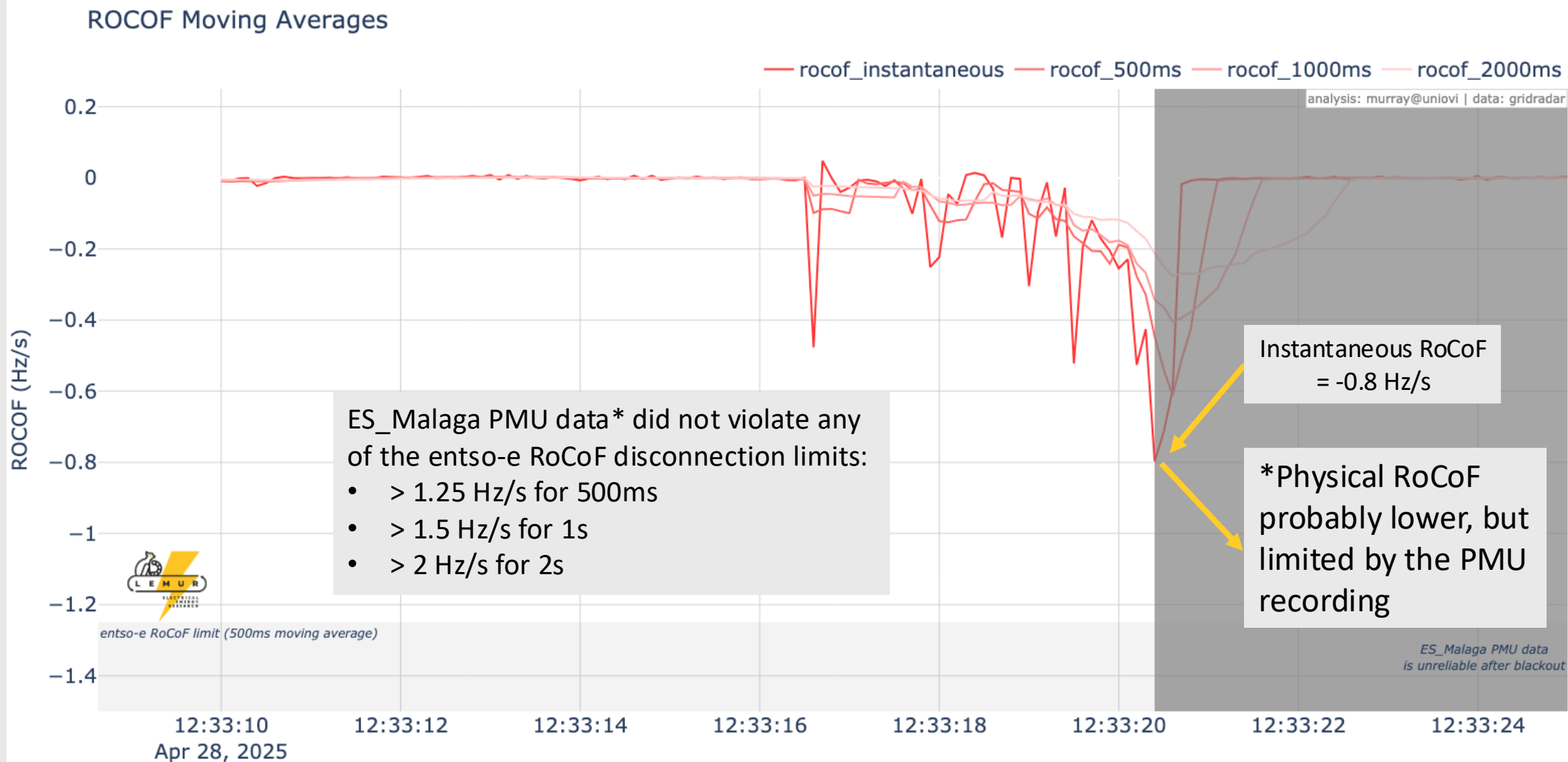
Rate of Change of Frequency

ROCOF OVERVIEW



ROCOF STAYED WITHIN LIMITS

RoCoF limits from "Rate of Change of Frequency (RoCoF withstand capability", entso-e, 31 January 2018

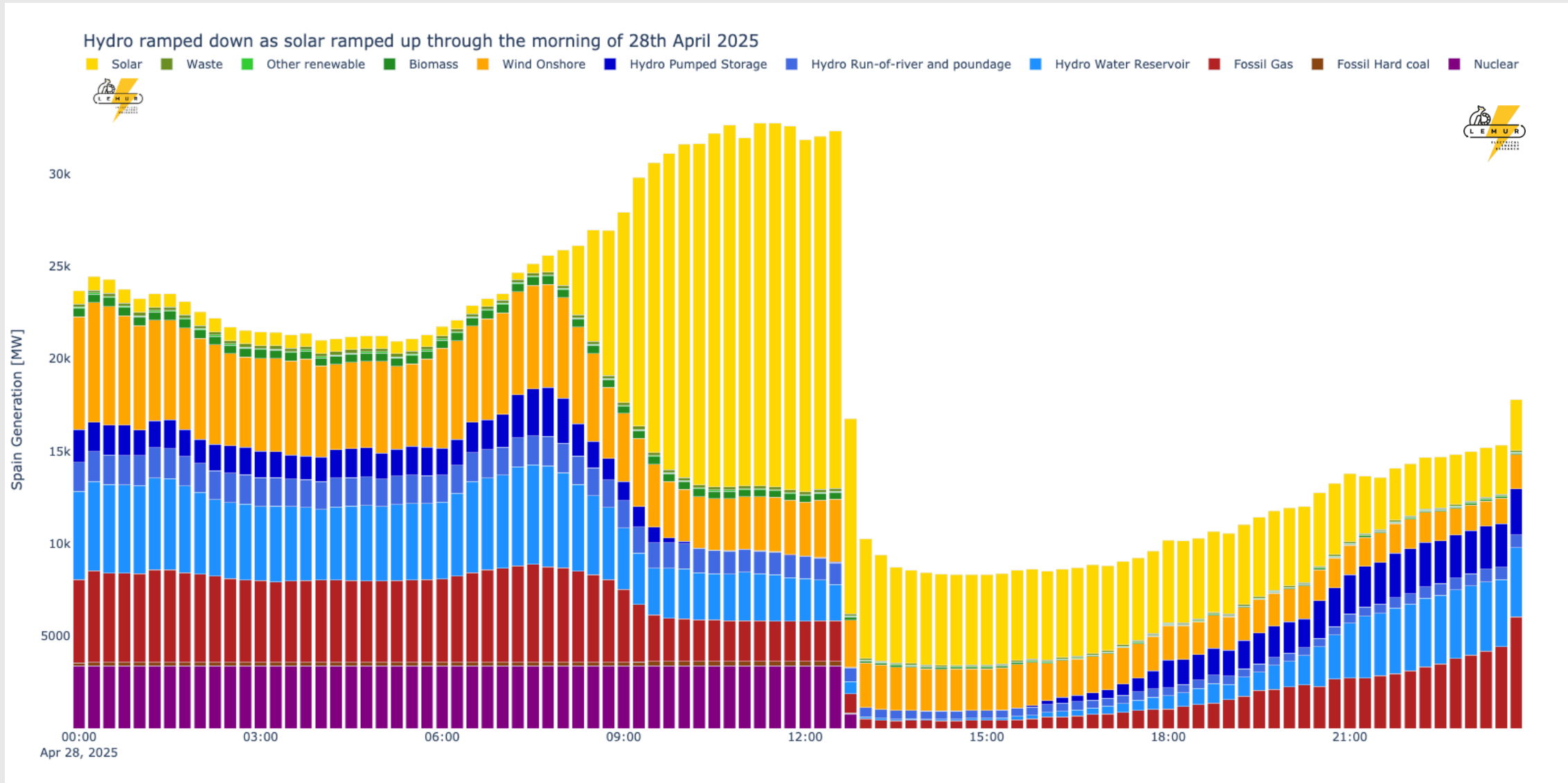




Section 3

Generation, Inertia, Exports

GENERATION MIX ON APRIL 28, 2025



Source: ENTSO-E Transparency Platform



INERTIA ESTIMATION METHODS – 1 OF 2

- Grid stability is a complex topic, but two very important factors are **controls** and **inertia**.
- **Controls:** Grid-Forming (GFM) controls allow inverter-based resources (“IBR”, e.g. solar, wind, batteries) to positively contribute to grid stability, but these are less prevalent today, and historically most IBRs use grid-following (GFL) controls. New battery installations will greatly expand available GFM assets.
- **Inertia:** [ENTSO-E “Project Inertia”](#) has published basic methods for estimating kinetic inertia, which we utilize in the following slides. These estimates should be considered lower bounds because they do not include:
 - Non-generating inertia contributors (e.g. synchronous compensators)
 - Inertia from online plants operating below nominal capacity
 - Contributions from GFM-capable inverters

INERTIA ESTIMATION METHODS – 2 OF 2

Lower bound estimate!

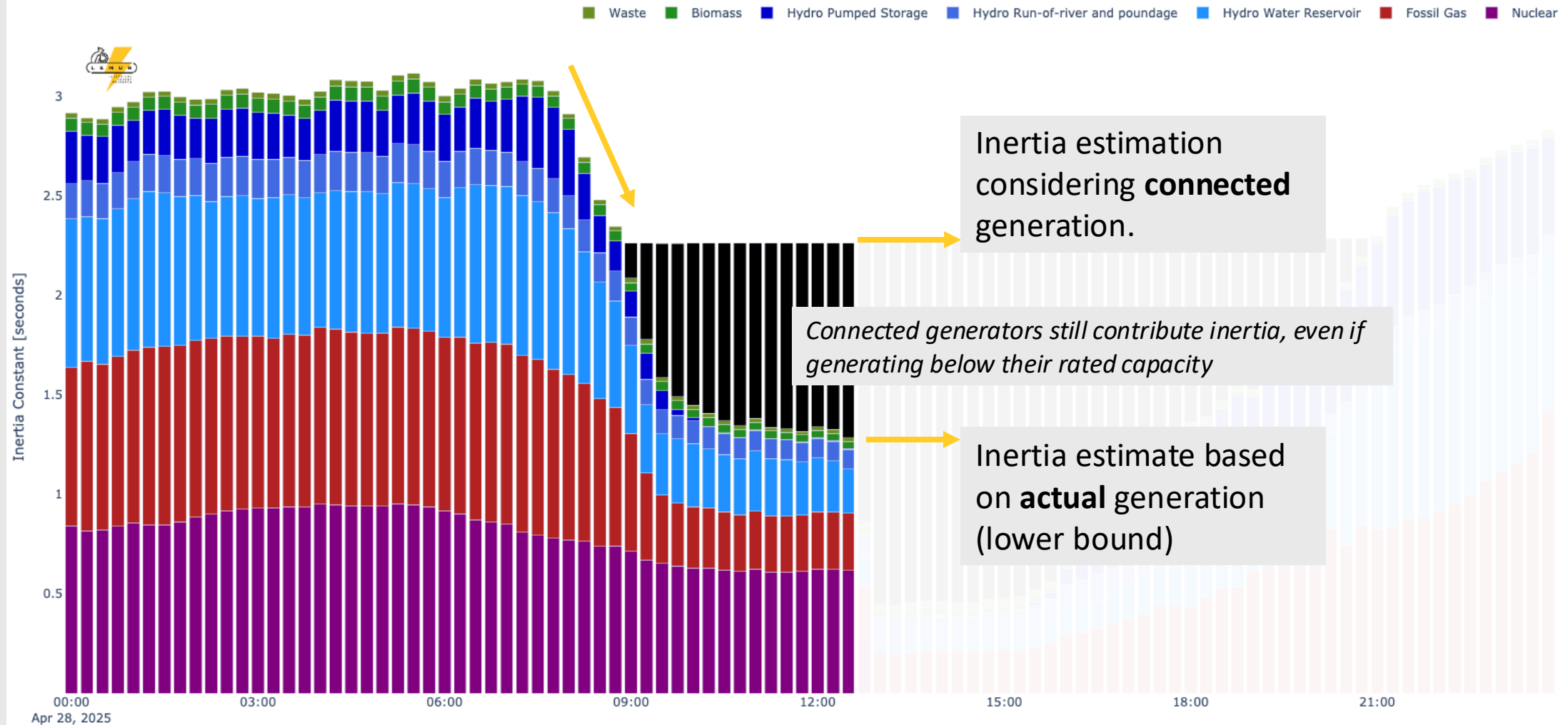


- The authors' lower-bound estimates are based on the simplest data available: generation [MW] by resource type at 12:30pm on April 28, 2025.
- Inertia estimates by resource type are simply the product of the that resource's share of the generation mix [GW/GW] and the inertia constants [s] published by ENTSO-E.
- Total equivalent inertia lower-bound estimates is simply the sum of these estimates.

Generation Type	Generation @ 12:30pm (MW)	H_eq (s)	Equivalent Inertia Floor (s)	Generation (%)	Inertia (%)
Solar	19340	0	0.00	59.8%	0.0%
Wind Onshore	3416	0	0.00	10.6%	0.0%
Nuclear	3384	5.9	0.62	10.5%	46.7%
Fossil Gas	2216	4.2	0.29	6.9%	21.8%
Hydro Water Reservoir	1952	3.7	0.22	6.0%	16.9%
Hydro Run-of-river and poundage	1184	2.7	0.10	3.7%	7.5%
Biomass	356	3.3	0.04	1.1%	2.7%
Fossil Hard coal	228	4.2	0.03	0.7%	2.2%
Waste	160	3.8	0.02	0.5%	1.4%
Other renewable	60	3.5	0.01	0.2%	0.5%
Hydro Pumped Storage	36	3.5	0.00	0.1%	0.3%
Total	32332		1.32	100.0%	100.0%

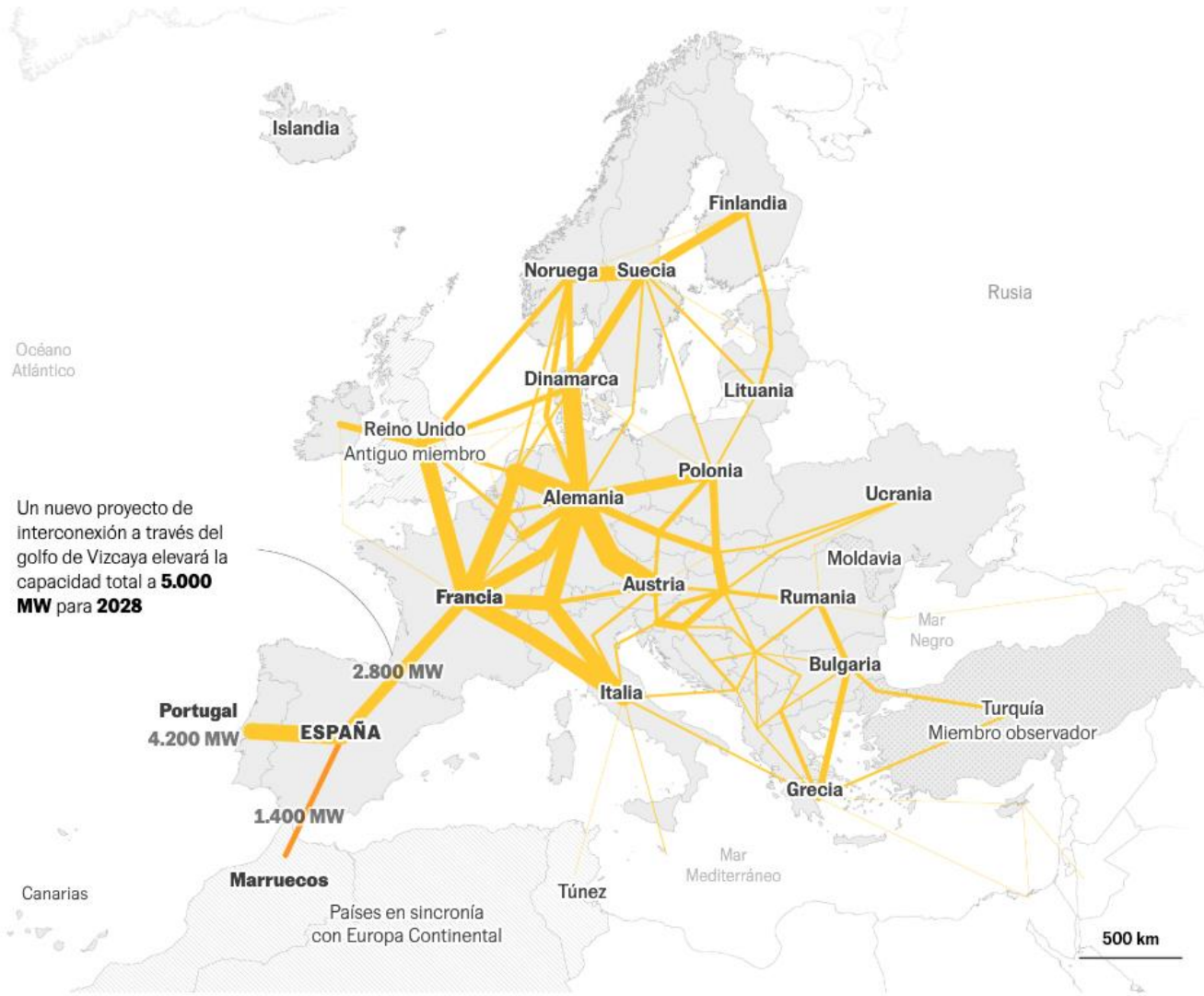
MINIMUM INERTIA ESTIMATES FOR APRIL 28, 2025

On 28th April 2025, the minimum system inertia can be estimated as the inertia constants weighted by share of the generation mix



La red eléctrica europea

- Capacidad de interconexión (MW)
- Países miembros de ENTSO-E



Fuente: Ember Energy, ENTSO-E, Red Eléctrica.

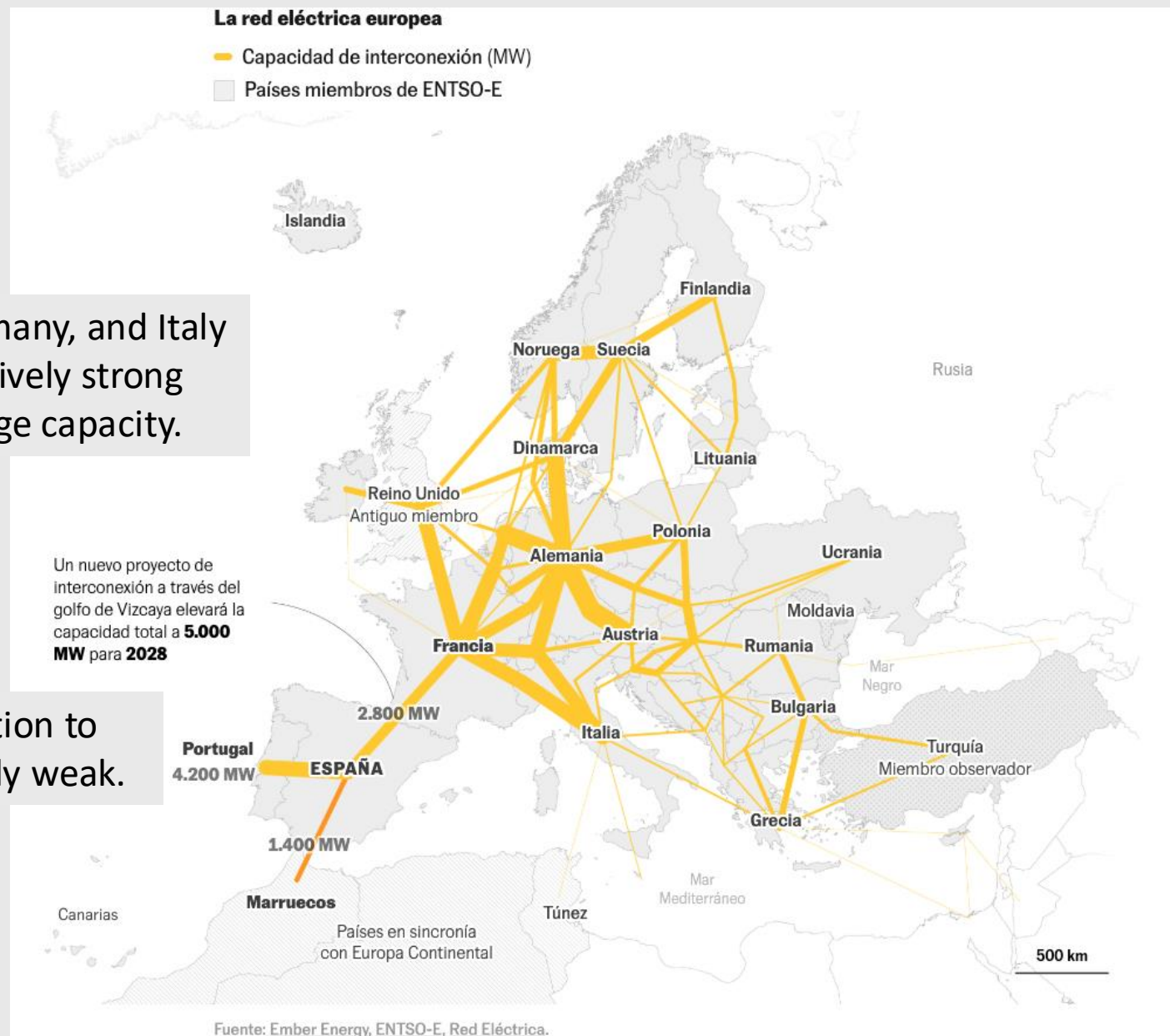
France, Germany, and Italy have relatively strong interchange capacity.

Iberian connection to France is relatively weak.

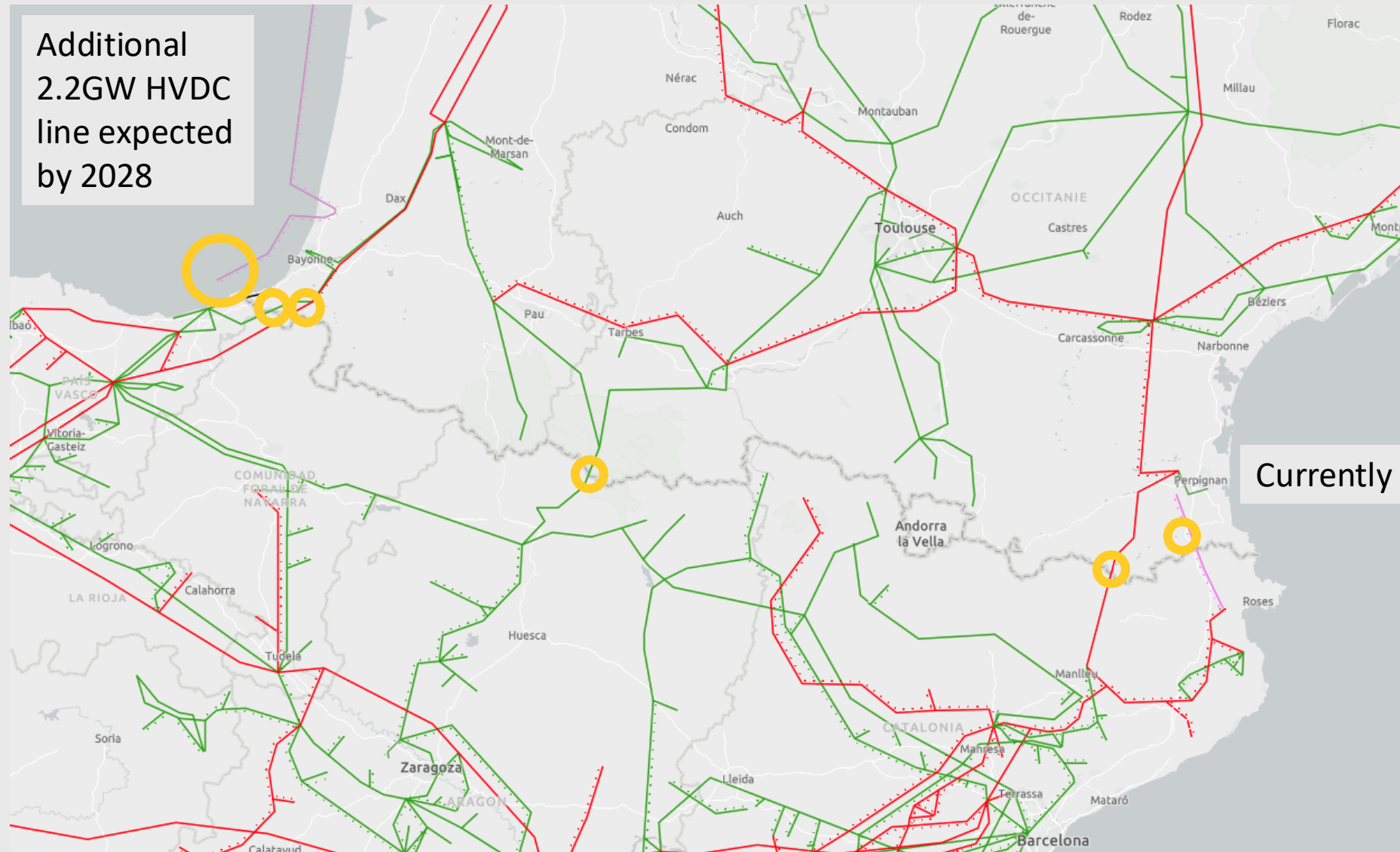
France, Germany, and Italy
have relatively strong
interchange capacity.

Un nuevo proyecto de
interconexión a través del
golfo de Vizcaya elevará la
capacidad total a **5.000
MW** para **2028**

Iberian connection to
France is relatively weak.

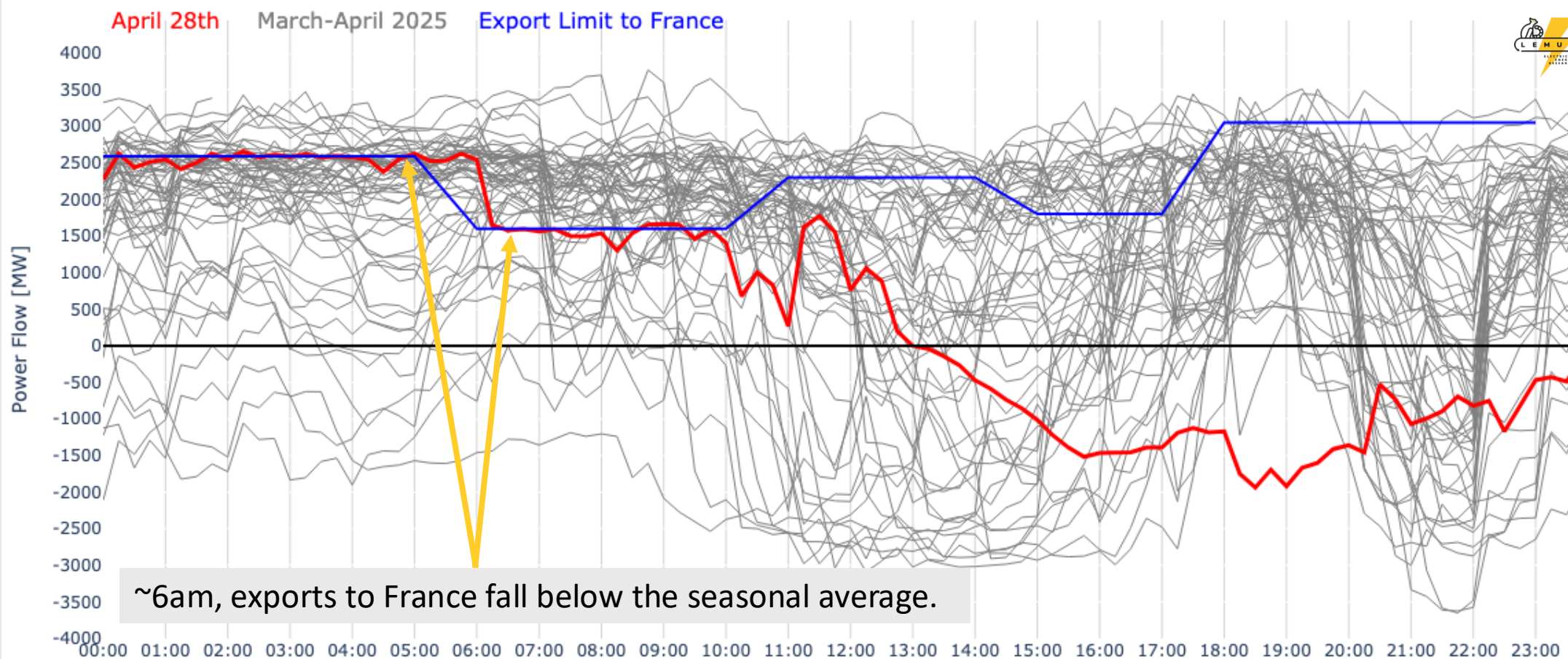


CONNECTIONS WITH FRANCE



EXPORTS TO FRANCE WERE BELOW AVERAGE

Exports from Spain to France were below seasonal average



Source: ENTSO-E Transparency Platform

Apagón – April 28, 2025



CONCLUSIONS

- On April 28th, 2025, the operating conditions on the Spanish grid were challenging, but not unprecedented.
- The authors are actively working to reconstruct the grid conditions of April 28 and will continue to publish our findings publicly.
- Any feedback or relevant data is appreciated!
 - Professor Pablo García Fernández: garciafpablo@uniovi.es
 - Bryan Murray: uo292264@uniovi.es



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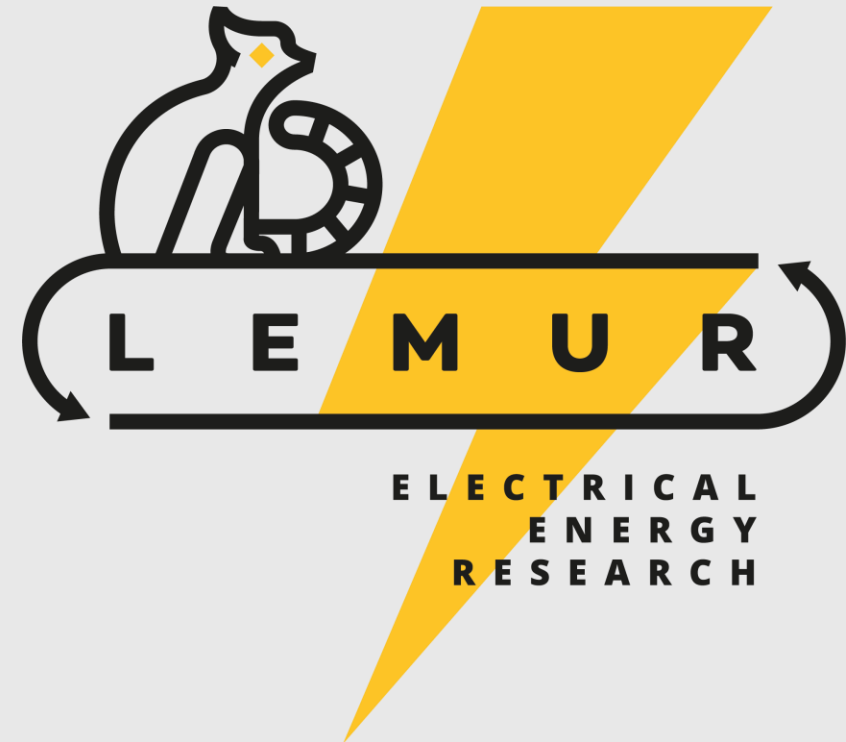


LEMUR



OUR GOAL

LEMUR is a **multidisciplinary research team** with researchers belonging to different knowledge areas: Electrical Engineering, Power Electronics and Automation and Control Engineering. The strength is based on looking for synergies and knowledge integration among team members in order to apply for bigger and complex research projects.



OUR VISION

To integrate in a laboratory an environment for testing of small scale distributed energy systems and their integration into the electrical grid. For the study and analysis, a four level approach is proposed:

- Development of generation systems and power converters for the injection of electrical energy into the grid under efficiency and reliability constraints.
- Development of energy storage systems for the compensation of transient demands at the electrical grid, including pulsating generation by renewable energy systems, grid contingencies and virtual zero demanding energy buildings.
- Development of coordinated control strategies for all the small scale generators installed at the grid in order the system to behave as a virtual integrated power plant. Development of new power flow algorithms for considering the unbalanced conditions at weak grids (single phase loads and generation systems).
- Study of new economic and infrastructure models relying on the distributed generation, which allow to maximize the benefits considering the existing and future regulations.



OUR RESEARCH LINES

ANALYSIS AND DESIGN OF POWER CONVERTER TOPOLOGIES FOR ENERGY STORAGE

- Design of power electronics converters with an emphasis of multiport topologies for energy storage applications

DYNAMIC ANALYSIS AND CONTROL OF POWER CONVERTERS FOR DISTRIBUTED RESOURCES

- High dynamic performance of power converters and microgrids by advanced control systems

POWER FLOW ANALYSIS FOR HYBRID DC/AC GRIDS WITH HIGH PENETRATION OF DISTRIBUTED RESOURCES

- Impact of the integration of distributed resources in the electrical grid

ENERGY DEMAND ANALYSIS AND VISUALIZATION

- Analytic tools and visualization strategies for energy planning

APPLICATIONS

- Industrial applications and knowledge transfer



THANK YOU!

Please get in touch!
